

**E9 Trigonometric functions
in context A level only**

Name: _____

Class: _____

Date: _____

Time: **467 minutes**

Marks: **387 marks**

Comments:

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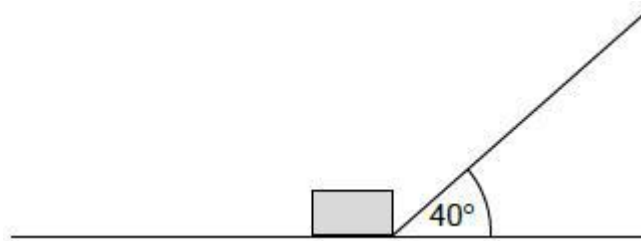
Q1.

In this question use $g = 9.8 \text{ m s}^{-2}$, giving your final answers to an appropriate degree of accuracy.

The diagram shows a box, of mass 8.0 kg , being pulled by a string so that the box moves at a constant speed along a rough horizontal wooden board.

The string is at an angle of 40° to the horizontal.

The tension in the string is 50 newtons .



The coefficient of friction between the box and the board is μ

Model the box as a particle.

- (a) Show that $\mu = 0.83$

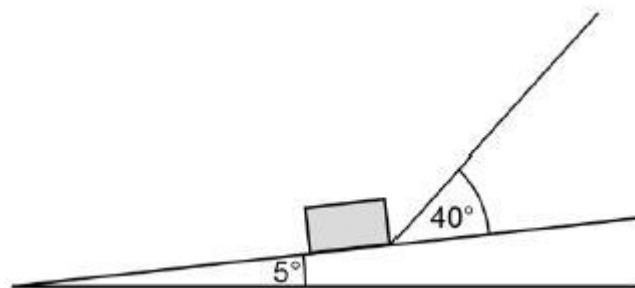
(4)

- (b) One end of the board is lifted up so that the board is now inclined at an angle of 5° to the horizontal.

The box is pulled up the inclined board.

The string remains at an angle of 40° to the board.

The tension in the string is increased so that the box accelerates up the board at 3 m s^{-2}



- (i) Draw a diagram to show the forces acting on the box as it moves.

(1)

- (ii) Find the tension in the string as the box accelerates up the slope at 3 m s^{-2} .

(7)

(Total 12 marks)

Q2.

- (a) Solve the differential equation

$$\frac{dx}{dt} = \frac{150 \cos 2t}{x}$$

given that $x = 20$ when $t = \frac{\pi}{4}$, giving your solution in the form $x^2 = f(t)$.

(6)

- (b) The oscillations of a 'baby bouncy cradle' are modelled by the differential equation

$$\frac{dx}{dt} = \frac{150 \cos 2t}{x}$$

where x cm is the height of the cradle above its base t seconds after the cradle begins to oscillate.

Given that the cradle is 20 cm above its base at time $t = \frac{\pi}{4}$ seconds, find:

- (i) the height of the cradle above its base 13 seconds after it starts oscillating, giving your answer to the nearest millimetre;
- (ii) the time at which the cradle will first be 11 cm above its base, giving your answer to the nearest tenth of a second.

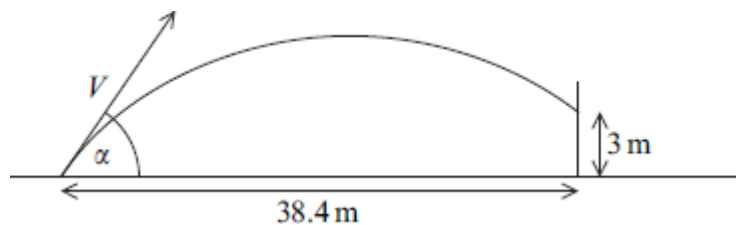
(2)

(2)

(Total 10 marks)

Q3.

A golf ball is hit from a point on a horizontal surface, so that it has an initial velocity V m s⁻¹ at an angle α above the horizontal. The ball travels through the air and after 2.4 seconds hits a vertical wall at a height of 3 metres. The wall is at a horizontal distance of 38.4 metres from the point where the ball was hit. The path of the ball is shown in the diagram.



Assume that the weight of the ball is the only force that acts on it as it travels through the air.

- (a) Find the horizontal component of the velocity of the ball.

(2)

(b) Find V .

(5)

(c) Find α .

(3)

(Total 10 marks)

Q4.

A particle moves in a horizontal plane. The vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors in the horizontal plane. At time t seconds, the velocity of the particle, $\mathbf{v}$ m s⁻¹, is given by

$$\mathbf{v} = 12 \cos \left(\frac{\pi}{3} t \right) \mathbf{i} - 9t^2 \mathbf{j}$$

(a) Find an expression for the acceleration of the particle at time t .

(2)

(b) The particle, which has mass 4 kg, moves under the action of a single force, $\mathbf{F}$ newtons.

(i) Find an expression for the force $\mathbf{F}$ in terms of t .

(2)

(ii) Find the magnitude of $\mathbf{F}$ when $t = 3$.

(2)

(c) When $t = 3$, the particle is at the point with position vector $(4\mathbf{i} - 2\mathbf{j})$ m.

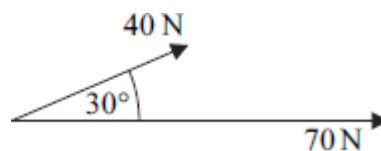
Find the position vector, $\mathbf{r}$ metres, of the particle at time t .

(5)

(Total 11 marks)

Q5.

Two forces, acting at a point, have magnitudes of 40 newtons and 70 newtons. The angle between the two forces is 30° , as shown in the diagram.



(a) Find the magnitude of the resultant of these two forces.

(4)

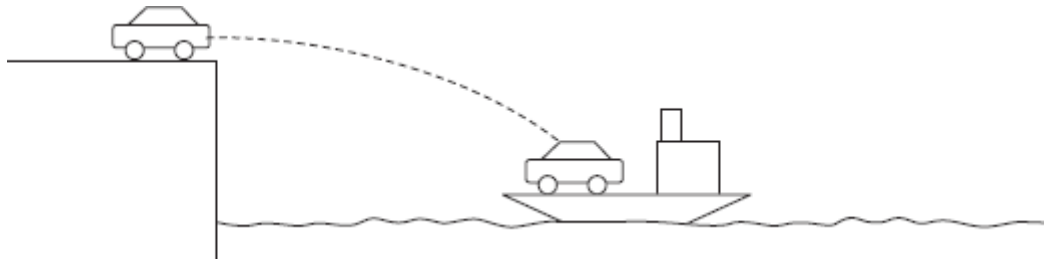
(b) Find the angle between the resultant force and the 70 newton force.

(3)

(Total 7 marks)

Q6.

In a scene from an action movie, a car is driven off the edge of a cliff and lands on the deck of a boat in the sea, as shown in the diagram.



To land on the boat, the car must move 20 metres horizontally from the cliff. The level of the deck of the boat is 8 metres below the top of the cliff. Assume that the car is a particle which is travelling horizontally when it leaves the top of the cliff and that the car is not affected by air resistance as it moves.

- (a) Find the time that it takes for the car to reach the deck of the boat. (3)
 - (b) Find the speed at which the car is travelling when it leaves the top of the cliff. (3)
 - (c) Find the speed of the car when it hits the deck of the boat. (4)
- (Total 10 marks)**

Q7.

A block of mass 30 kg is dragged across a rough horizontal surface by a rope that is at an angle of 20° to the horizontal. The coefficient of friction between the block and the surface is 0.4.

- (a) The tension in the rope is 150 newtons.
 - (i) Draw a diagram to show the forces acting on the block as it moves. (2)
 - (ii) Show that the magnitude of the normal reaction force on the block is 243 newtons, correct to three significant figures. (3)
 - (iii) Find the magnitude of the friction force acting on the block. (2)
 - (iv) Find the acceleration of the block. (4)
- (b) When the block is moving, the tension is reduced so that the block moves at a constant speed, with the angle between the rope and the horizontal unchanged. Find the tension in the rope when the block is moving at this constant speed. (5)

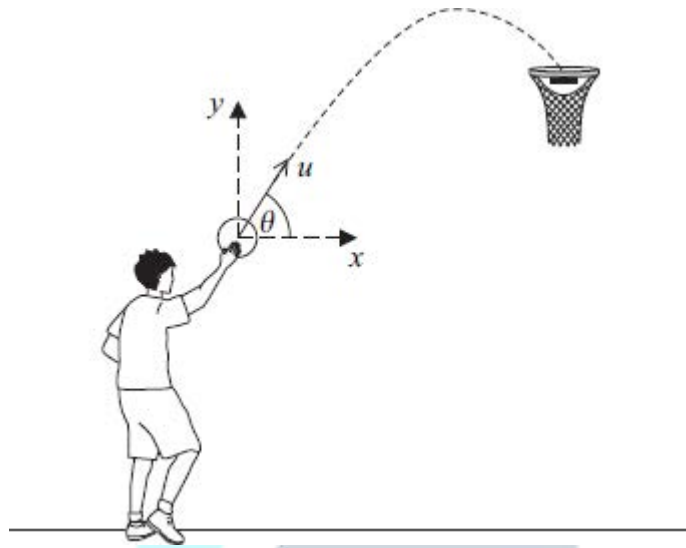
- (c) If the block were made to move at a greater **constant** speed, again with the angle between the rope and the horizontal unchanged, how would the tension in this case compare to the tension found in part (b)?

(1)

(Total 17 marks)

Q8.

A player projects a basketball with speed $u \text{ m s}^{-1}$ at an angle θ above the horizontal. The basketball travels in a vertical plane through the point of projection and goes into the basket. During the motion, the horizontal and upward vertical displacements of the basketball from the point of projection are x metres and y metres respectively.



- (a) Find an expression for y in terms of x , u , g and $\tan \theta$.

(6)

- (b) The player projects the basketball with speed 8 m s^{-1} from a point 0.5 metres vertically below and 5 metres horizontally from the basket.

- (i) Show that the two possible values of θ are approximately 63.1° and 32.6° , correct to three significant figures.

(5)

- (ii) Given that the player projects the basketball at 63.1° to the horizontal, find the direction of the motion of the basketball as it enters the basket. Give your answer to the nearest degree.

(4)

- (c) State a modelling assumption needed for answering parts (a) and (b) of this question.

(1)

(Total 16 marks)

Q9.

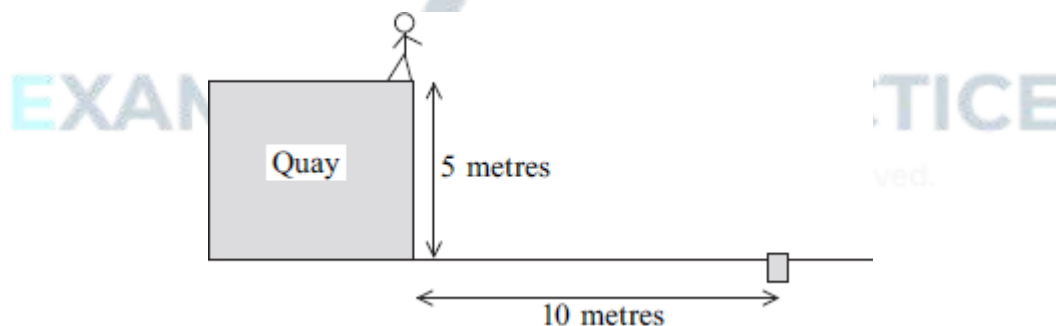
A cyclist freewheels, with a constant acceleration, in a straight line down a slope.

As the cyclist moves 50 metres, his speed increases from 4 m s^{-1} to 10 m s^{-1} .

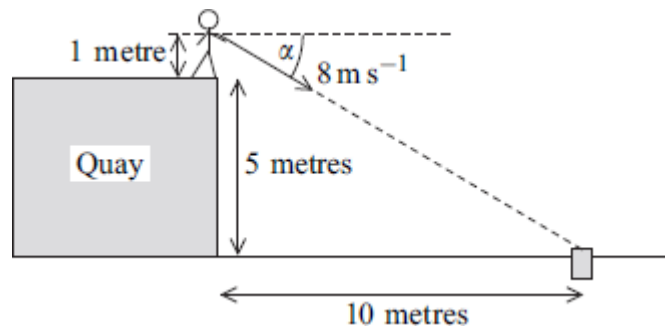
- (a) (i) Find the acceleration of the cyclist. (3)
- (ii) Find the time that it takes the cyclist to travel this distance. (3)
- (b) The cyclist has a mass of 70 kg. Calculate the magnitude of the resultant force acting on the cyclist. (2)
- (c) The slope is inclined at an angle α to the horizontal.
- (i) Find α if it is assumed that there is no resistance force acting on the cyclist. (3)
- (ii) Find α if it is assumed that there is a constant resistance force of magnitude 30 newtons acting on the cyclist. (3)
- (d) Make a criticism of the assumption described in part (c)(ii). (1)
- (Total 15 marks)**

Q10.

A girl stands at the edge of a quay and sees a tin can floating in the water. The water level is 5 metres below the top of the quay and the can is at a horizontal distance of 10 metres from the quay, as shown in the diagram.



The girl decides to throw a stone at the can. She throws the stone from a height of 1 metre above the top of the quay. The initial velocity of the stone is 8 m s^{-1} at an angle α below the horizontal, so that the initial velocity of the stone is directed at the can, as shown in the diagram.



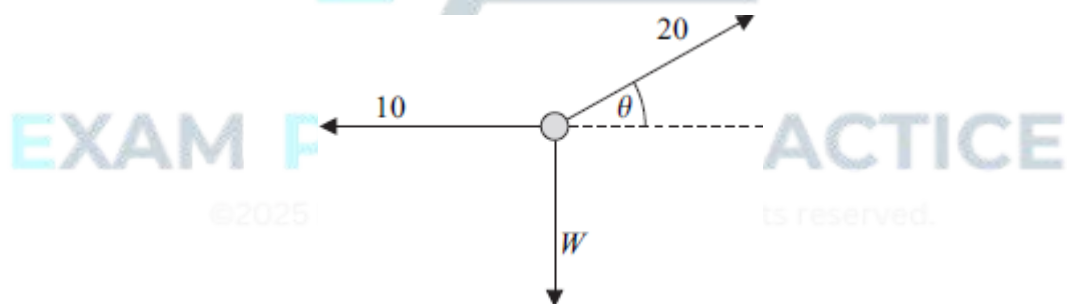
Assume that the stone is a particle and that it experiences no air resistance as it moves.

- (a) Find α . (2)
- (b) Find the time that it takes for the stone to reach the level of the water. (6)
- (c) Find the distance between the stone and the can when the stone hits the water. (4)

(Total 12 marks)

Q11.

A particle, of weight W newtons, is held in equilibrium by two forces of magnitudes 10 newtons and 20 newtons. The 10-newton force is horizontal and the 20-newton force acts at an angle θ above the horizontal, as shown in the diagram. All three forces act in the same vertical plane.



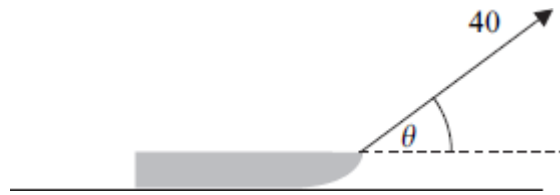
- (a) Find θ . (3)
- (b) Find W . (2)
- (c) Calculate the mass of the particle. (2)

(Total 7 marks)

Q12.

A child pulls a sledge, of mass 8 kg, along a rough horizontal surface, using a light rope.

The coefficient of friction between the sledge and the surface is 0.3. The tension in the rope is 40 newtons. The rope is kept at an angle of θ to the horizontal, as shown in the diagram.



Model the sledge as a particle.

- (a) Draw a diagram to show all the forces acting on the sledge. (1)
- (b) Find the magnitude of the normal reaction force acting on the sledge, in terms of θ . (3)
- (c) The acceleration of the sledge is

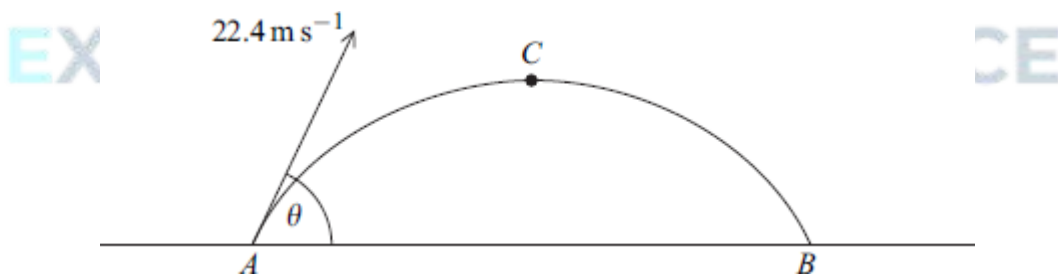
$$p + q \cos \theta + r \sin \theta$$

Find p , q and r .

(5)
(Total 9 marks)

Q13.

A particle is launched from the point A on a horizontal surface, with a velocity of 22.4 m s^{-1} at an angle θ above the horizontal, as shown in the diagram.



After 2 seconds, the particle reaches the point C , where it is at its maximum height above the surface.

- (a) Show that $\sin \theta = 0.875$. (3)
- (b) Find the height of the point C above the horizontal surface. (3)
- (c) The particle returns to the surface at the point B . Find the distance between A and B .

(3)

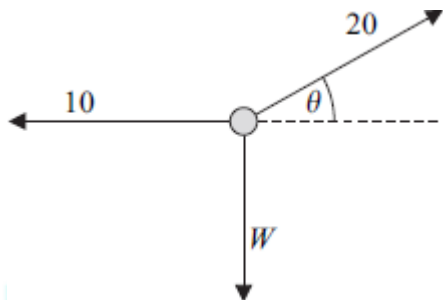
- (d) Find the length of time during which the height of the particle above the surface is greater than 5 metres.

(5)

(Total 14 marks)

Q14.

A particle, of weight W newtons, is held in equilibrium by two forces of magnitudes 10 newtons and 20 newtons. The 10-newton force is horizontal and the 20-newton force acts at an angle θ above the horizontal, as shown in the diagram. All three forces act in the same vertical plane.



- (a) Find θ .

(3)

- (b) Find W .

(2)

- (c) Calculate the mass of the particle.

(2)

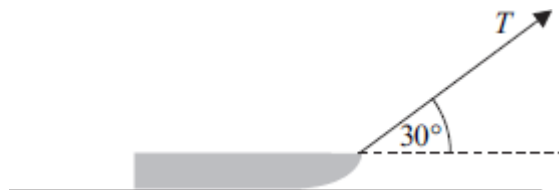
(Total 7 marks)

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Q15.

A child pulls a sledge, of mass 8 kg, along a rough horizontal surface, using a light rope. The coefficient of friction between the sledge and the surface is 0.3. The tension in the rope is T newtons. The rope is kept at an angle of 30° to the horizontal, as shown in the diagram.



Model the sledge as a particle.

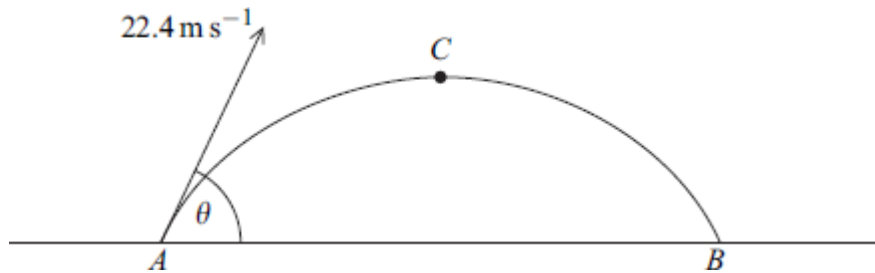
- (a) Draw a diagram to show all the forces acting on the sledge.

(1)

- (b) Find the magnitude of the normal reaction force acting on the sledge, in terms of T . (3)
- (c) Given that the sledge accelerates at 0.05 m s^{-2} , find T . (6)
- (Total 10 marks)

Q16.

A particle is launched from the point A on a horizontal surface, with a velocity of 22.4 m s^{-1} at an angle θ above the horizontal, as shown in the diagram.



After 2 seconds, the particle reaches the point C , where it is at its maximum height above the surface.

- (a) Show that $\sin \theta = 0.875$. (3)
- (b) Find the height of the point C above the horizontal surface. (3)
- (c) The particle returns to the surface at the point B . Find the distance between A and B . (3)
- (d) Find the length of time during which the height of the particle above the surface is greater than 5 metres. (5)
- (e) Find the minimum speed of the particle. (2)

(Total 16 marks)

Q17.

(In this question, take $g = 10 \text{ m s}^{-2}$.)

A projectile is fired from a point O with speed u at an angle of elevation α above the horizontal so as to pass through a point P . The projectile travels in a vertical plane through O and P . The point P is at a horizontal distance $2k$ from O and at a vertical distance k above O .

- (a) Show that α satisfies the equation

$$20k \tan^2 \alpha - 2u^2 \tan \alpha + u^2 + 20k = 0$$

(7)

- (b) Deduce that

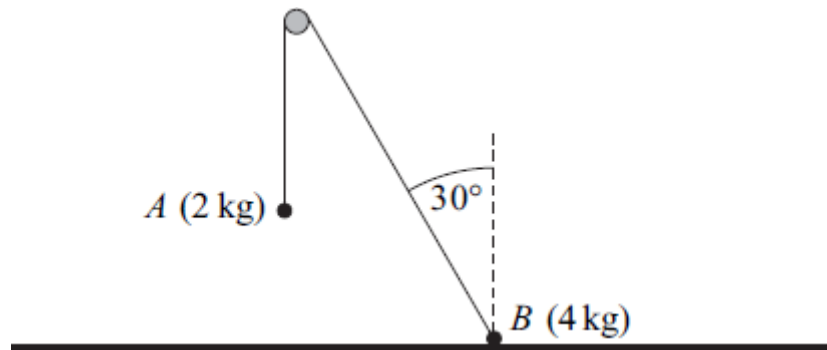
$$u^4 - 20ku^2 - 400k^2 \geq 0$$

(3)

(Total 10 marks)

Q18.

Two particles, A and B , are connected by a light inextensible string which passes over a smooth peg. Particle A has mass 2 kg and particle B has mass 4 kg. Particle A hangs freely with the string vertical. Particle B is at rest in equilibrium on a rough horizontal surface with the string at an angle of 30° to the vertical. The particles, peg and string are shown in the diagram.



- (a) By considering particle A , find the tension in the string.

(2)

- (b) Draw a diagram to show the forces acting on particle B .

(2)

- (c) Show that the magnitude of the normal reaction force acting on particle B is 22.2 newtons, correct to three significant figures.

(3)

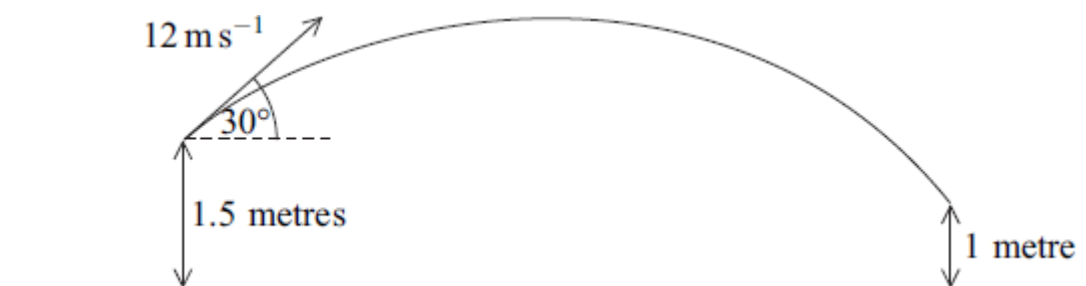
- (d) Find the least possible value of the coefficient of friction between particle B and the surface.

(4)

(Total 11 marks)

Q19.

An arrow is fired from a point at a height of 1.5 metres above horizontal ground. It has an initial velocity of 12 m s^{-1} at an angle of 30° above the horizontal. The arrow hits a target at a height of 1 metre above horizontal ground. The path of the arrow is shown in the diagram.

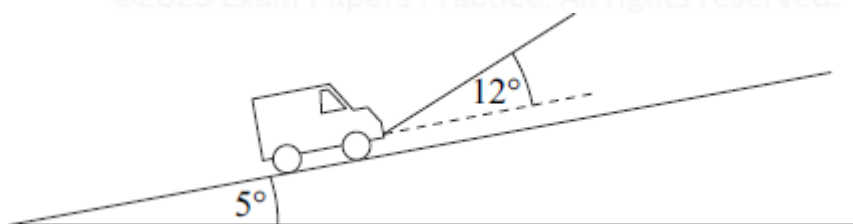


Model the arrow as a particle.

- (a) Show that the time taken for the arrow to travel to the target is 1.30 seconds, correct to three significant figures. (5)
 - (b) Find the horizontal distance between the point where the arrow is fired and the target. (2)
 - (c) Find the speed of the arrow when it hits the target. (4)
 - (d) Find the angle between the velocity of the arrow and the horizontal when the arrow hits the target. (2)
 - (e) State one assumption that you have made about the forces acting on the arrow. (1)
- (Total 14 marks)**

Q20.

A van, of mass 2000 kg, is towed up a slope inclined at 5° to the horizontal. The tow rope is at an angle of 12° to the slope. The motion of the van is opposed by a resistance force of magnitude 500 newtons. The van is accelerating up the slope at 0.6 m s^{-2} .

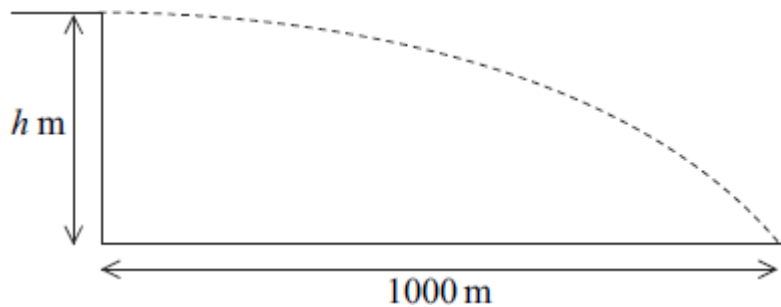


Model the van as a particle.

- (a) Draw a diagram to show the forces acting on the van. (2)
 - (b) Show that the tension in the tow rope is 3480 newtons, correct to three significant figures. (5)
- (Total 7 marks)**

Q21.

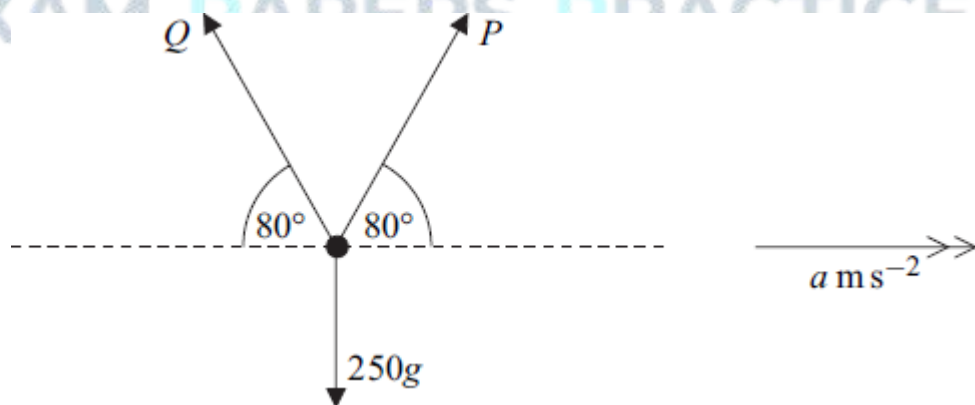
A bullet is fired horizontally from the top of a vertical cliff, at a height of h metres above the sea. It hits the sea 4 seconds after being fired, at a distance of 1000 metres from the base of the cliff, as shown in the diagram.



- (a) Find the initial speed of the bullet. (2)
 - (b) Find h . (2)
 - (c) Find the speed of the bullet when it hits the sea. (4)
 - (d) Find the angle between the velocity of the bullet and the horizontal when it hits the sea. (3)
- (Total 11 marks)**

Q22.

Three forces act in a vertical plane on an object of mass 250 kg, as shown in the diagram.



The two forces P newtons and Q newtons each act at 80° to the horizontal. The object accelerates horizontally at $a \text{ m s}^{-2}$ under the action of these forces.

- (a) Show that

$$P = 125 \left(\frac{a}{\cos 80^\circ} + \frac{g}{\sin 80^\circ} \right)$$

(5)

- (b) Find the value of a for which Q is zero.

(3)

(Total 8 marks)

Q23.

(In this question, use $g = 10 \text{ m s}^{-2}$.)

A golf ball is hit from a point O on a horizontal golf course with a velocity of 40 m s^{-1} at an angle of elevation θ . The golf ball travels in a vertical plane through O . During its flight, the horizontal and upward vertical distances of the golf ball from O are x and y metres respectively.

- (a) Show that the equation of the trajectory of the golf ball during its flight is given by

$$x^2 \tan^2 \theta - 320x \tan \theta + (x^2 + 320y) = 0$$

(6)

- (b) (i) The golf ball hits the top of a tree, which has a vertical height of 8 m and is at a horizontal distance of 150 m from O .

Find the two possible values of θ .

(5)

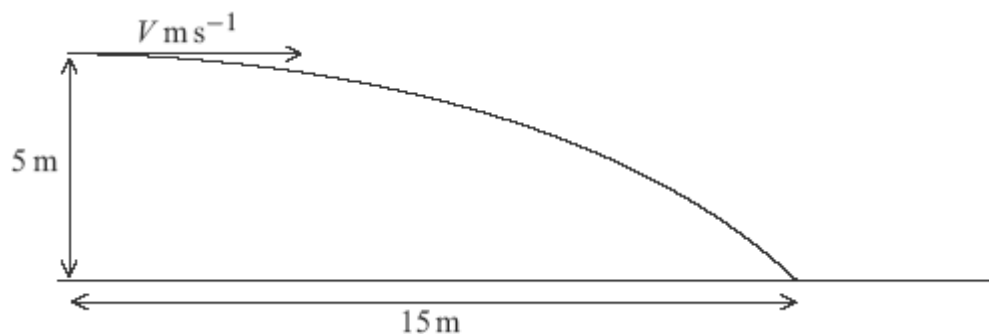
- (ii) Which value of θ gives the shortest possible time for the golf ball to travel from O to the top of the tree? Give a reason for your choice of θ .

(2)

(Total 13 marks)

Q24.

A ball is projected horizontally with speed $V \text{ m s}^{-1}$ at a height of 5 metres above horizontal ground. When the ball has travelled a horizontal distance of 15 metres, it hits the ground.



- (a) Show that the time it takes for the ball to travel to the point where it hits the ground is 1.01 seconds, correct to three significant figures.

(3)

- (b) Find V .

(2)

- (c) Find the speed of the ball when it hits the ground.

(4)

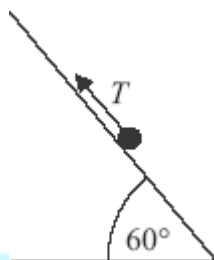
- (d) Find the angle between the velocity of the ball and the horizontal when the ball hits the ground. Give your answer to the nearest degree.

(3)

(Total 12 marks)

Q25.

A particle of mass 3 kg is on a smooth slope inclined at 60° to the horizontal. The particle is held at rest by a force of T newtons parallel to the slope, as shown in the diagram.



- (a) Draw a diagram to show all the forces acting on the particle.

(1)

- (b) Show that the magnitude of the normal reaction acting on the particle is 14.7 newtons.

(2)

- (c) Find T .

(2)

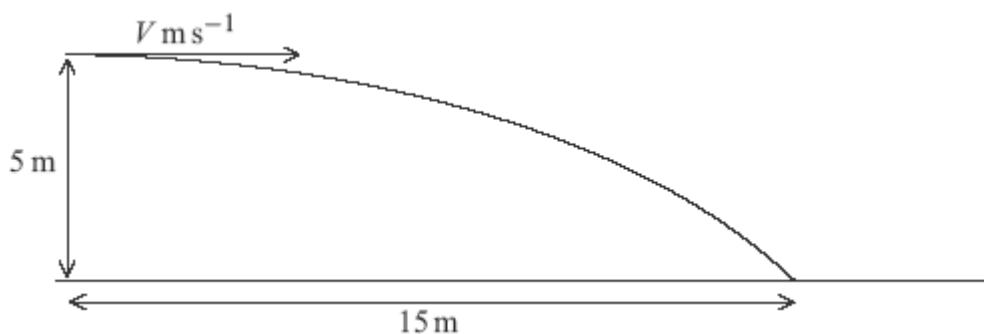
(Total 5 marks)

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Q26.

A ball is projected horizontally with speed $V \text{ m s}^{-1}$ at a height of 5 metres above horizontal ground. When the ball has travelled a horizontal distance of 15 metres, it hits the ground.



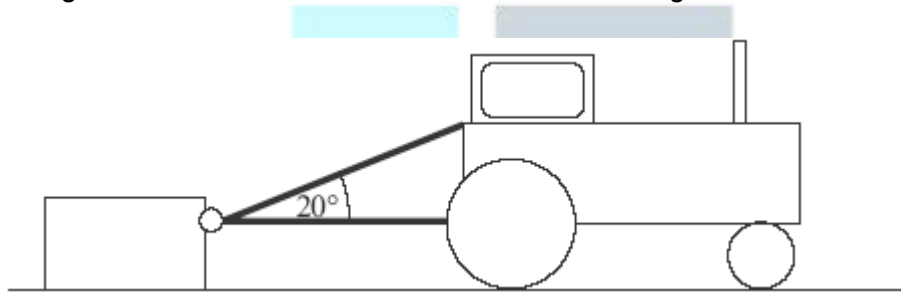
- (a) Show that the time it takes for the ball to travel to the point where it hits the ground is 1.01 seconds, correct to three significant figures.

- (3)
- (b) Find V . (2)
- (c) Find the speed of the ball when it hits the ground. (4)
- (d) Find the angle between the velocity of the ball and the horizontal when the ball hits the ground. Give your answer to the nearest degree. (3)
- (e) State two assumptions that you have made about the ball while it is moving. (2)

(Total 14 marks)

Q27.

A crate, of mass 200 kg, is initially at rest on a rough horizontal surface. A smooth ring is attached to the crate. A light inextensible rope is passed through the ring, and each end of the rope is attached to a tractor. The lower part of the rope is horizontal and the upper part is at an angle of 20° to the horizontal, as shown in the diagram.



When the tractor moves forward, the crate accelerates at 0.3 m s^{-2} . The coefficient of friction between the crate and the surface is 0.4.

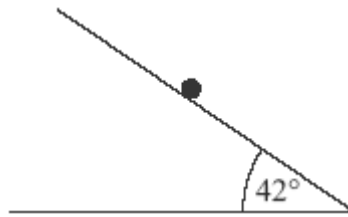
Assume that the tension, T newtons, is the same in both parts of the rope.

- (a) Draw and label a diagram to show the forces acting on the crate. (2)
- (b) Express the normal reaction between the surface and the crate in terms of T . (3)
- (c) Find T . (5)

(Total 10 marks)

Q28.

A particle, of mass m kg, is at rest on a rough plane which is inclined at an angle of 42° to the horizontal, as shown in the diagram.



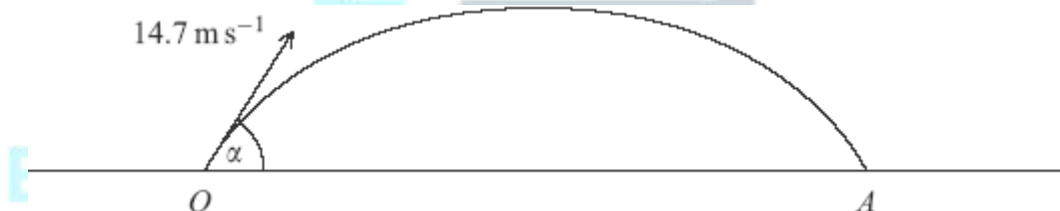
The friction force acting on the particle has magnitude 30 newtons.

- (a) Draw and label a diagram to show all the forces acting on the particle. (1)
- (b) Find m . (3)
- (c) Find the magnitude of the normal reaction force acting on the particle. (2)
- (d) Given that the particle is on the point of sliding down the plane, find the coefficient of friction between the particle and the plane. (3)

(Total 9 marks)

Q29.

A ball is struck so that it leaves a horizontal surface travelling at 14.7 m s^{-1} at an angle α above the horizontal. The path of the ball is shown in the diagram.



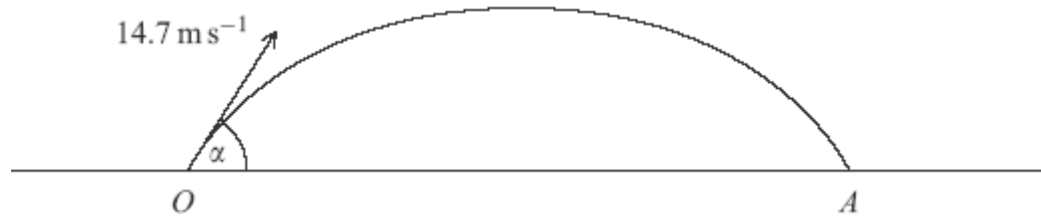
- (a) Show that the ball takes $\frac{3 \sin \alpha}{2}$ seconds to reach its maximum height. (3)
- (b) The ball reaches a maximum height of 7 metres.
 - (i) Find α . (5)
 - (ii) Find the range, OA . (3)

(Total 11 marks)

Q30.

A ball is struck so that it leaves a horizontal surface travelling at 14.7 m s^{-1} at an angle α

above the horizontal. The path of the ball is shown in the diagram.



- (a) Show that the ball takes $\frac{3 \sin \alpha}{2}$ seconds to reach its maximum height. (3)
- (b) The ball reaches a maximum height of 7 metres.
- (i) Find α . (5)
- (ii) Find the range, OA . (3)
- (c) State two assumptions that you needed to make in order to answer the earlier parts of this question. (2)
- (Total 13 marks)

Q31.

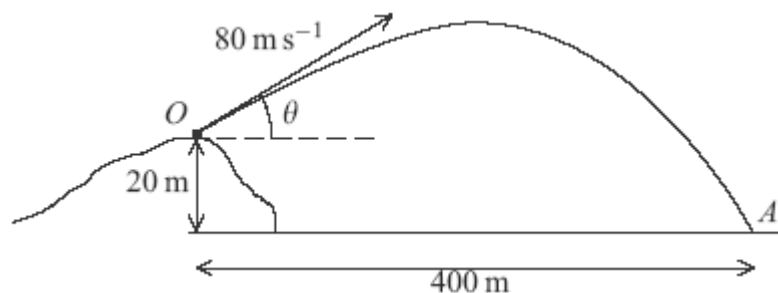
A projectile is fired from a point O on top of a hill with initial velocity 80 m s^{-1} at an angle θ above the horizontal and moves in a vertical plane. The horizontal and upward vertical distances of the projectile from O are x metres and y metres respectively.

- (a) (i) Show that, during the flight, the equation of the trajectory of the projectile is given by

$$y = x \tan \theta - \frac{gx^2}{12800} (1 + \tan^2 \theta)$$

(5)

- (ii) The projectile hits a target A , which is 20 m vertically below O and 400 m horizontally from O .



Show that

$$49 \tan^2 \theta - 160 \tan \theta + 41 = 0 \quad (2)$$

- (b) (i) Find the two possible values of θ . Give your answers to the nearest 0.1° . (3)
- (ii) Hence find the shortest possible time of the flight of the projectile from O to A . (2)
- (c) State a necessary modelling assumption for answering part (a) (i). (1)

(Total 13 marks)

Q32.

A football is kicked from ground level with an initial velocity of 30 m s^{-1} at an angle of 35° above the horizontal.

- (a) Find the maximum height of the ball above ground level. (4)
- (b) Show that when the speed of the ball is 28 m s^{-1} , the magnitude of the vertical component of its velocity is 13.4 m s^{-1} , correct to three significant figures. (4)
- (c) Find the angle between the velocity of the ball and the horizontal when the speed of the ball is 28 m s^{-1} . (2)

(Total 10 marks)

Q33.

A sledge of mass 8 kg is at rest on a rough horizontal surface. A child tries to move the sledge by pushing it with a pole, as shown in the diagram, but the sledge **does not move**. The pole is at an angle of 30° to the horizontal and exerts a force of 40 newtons on the sledge.



Model the sledge as a particle.

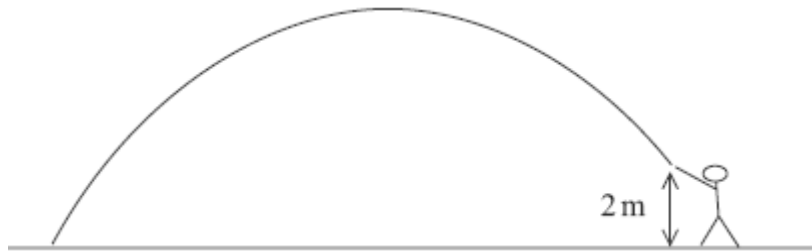
- (a) Draw a diagram to show the four forces acting on the sledge. (1)
- (b) Show that the normal reaction force between the sledge and the surface has magnitude 98.4 N . (3)

- (c) Find the magnitude of the friction force that acts on the sledge. (2)
- (d) Find the least possible value of the coefficient of friction between the sledge and the surface. (3)
- (Total 9 marks)

Q34.

A cricket ball is hit at ground level on a horizontal surface. It initially moves at 28 m s^{-1} at an angle of 50° above the horizontal.

- (a) Find the maximum height of the ball during its flight. (4)
- (b) The ball is caught when it is at a height of 2 metres above ground level, as shown in the diagram.

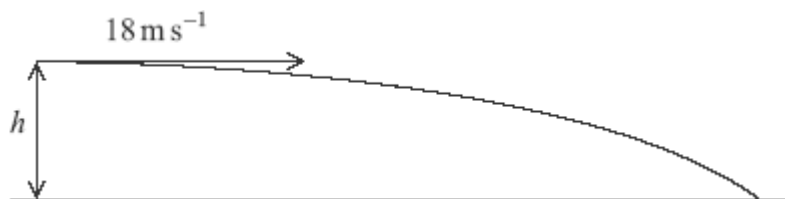


Show that the time that it takes for the ball to travel from the point where it was hit to the point where it was caught is 4.28 seconds, correct to three significant figures.

- (c) Find the speed of the ball when it is caught. (5)
- (Total 14 marks)

Q35.

An arrow is fired horizontally at a speed of 18 m s^{-1} from a point at a height of h metres above horizontal ground. The arrow hits the ground after it has been moving for 0.6 seconds. Model the arrow as a particle that moves only under the influence of gravity.



- (a) Show that $h = 1.76$, correct to three significant figures. (2)
- (b) Find the horizontal distance travelled by the arrow during its flight.

(2)

- (c) Find the speed of the arrow and the direction in which it is moving when it hits the ground.

(6)

(Total 10 marks)



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